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(54) **VEHICLE SEAT ADJUSTMENT ASSEMBLY**

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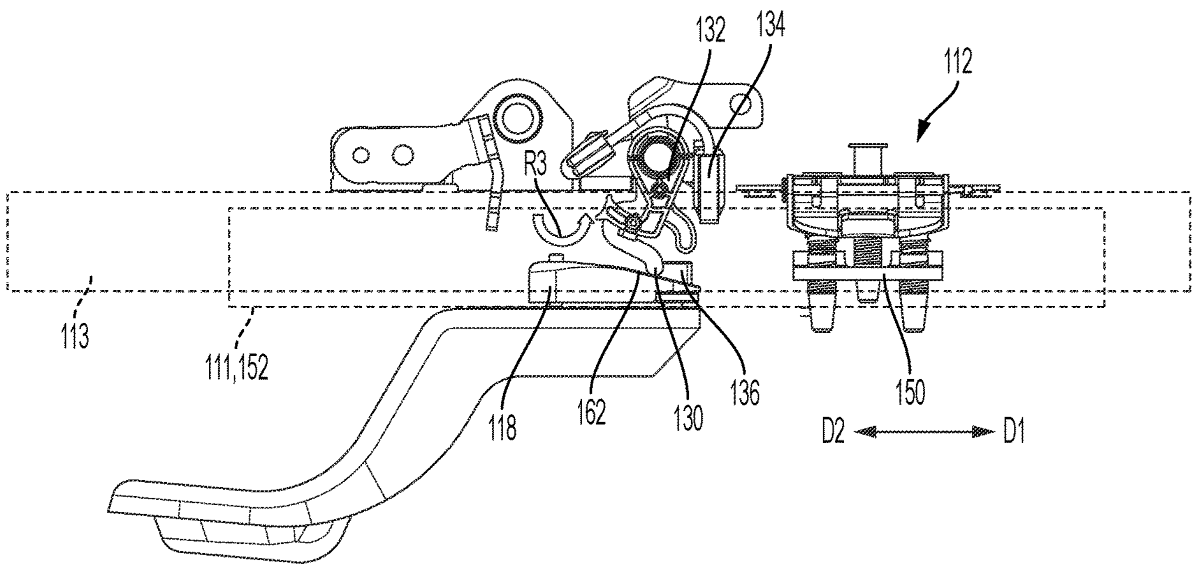
(57) **ABSTRACT**

A memory bracket configured for use in a vehicle seat rail assembly. The rail assembly including a translatable rail provided with a first stop member and the memory bracket fixed to a fixed rail and including a second stop member. In response to the translatable rail being positioned in a use position, the first stop member contacts the second stop member, and as the translatable rail moves in the longitudinal direction from an ingress-egress position to the one or more use positions, a portion of the memory bracket engages and displaces a mechanical effector of an ingress-egress aid to permit the backrest to pivot away from the seating surface.



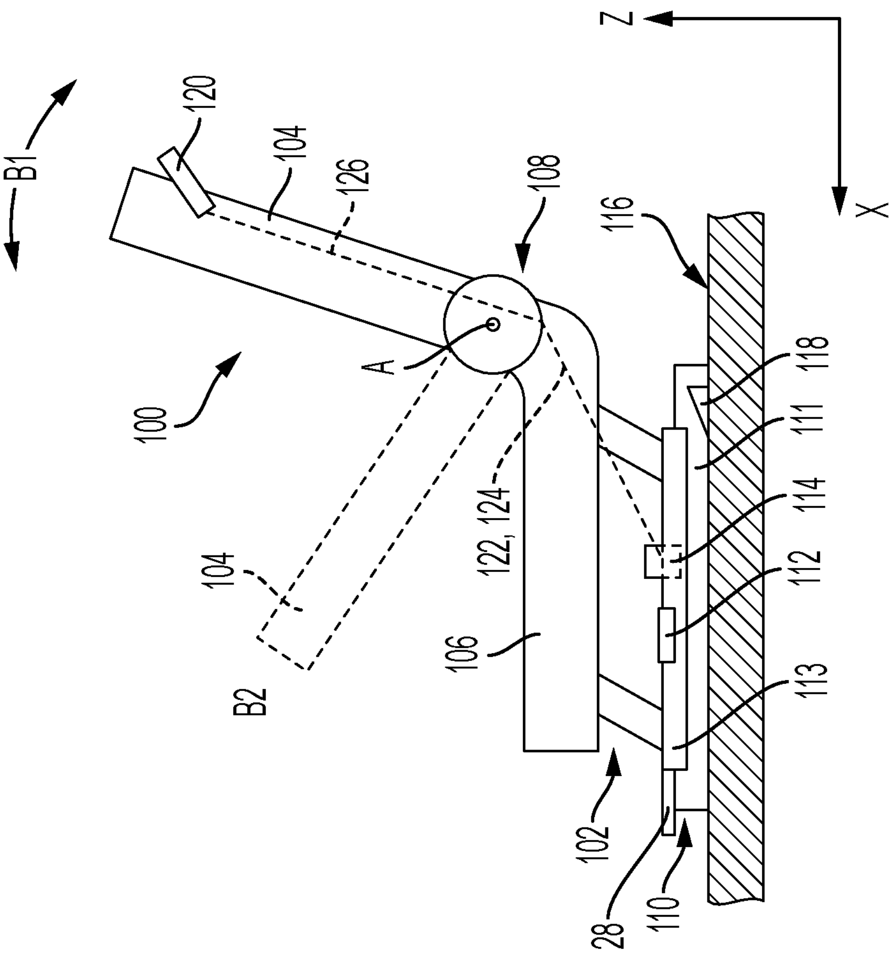

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FIG. 1

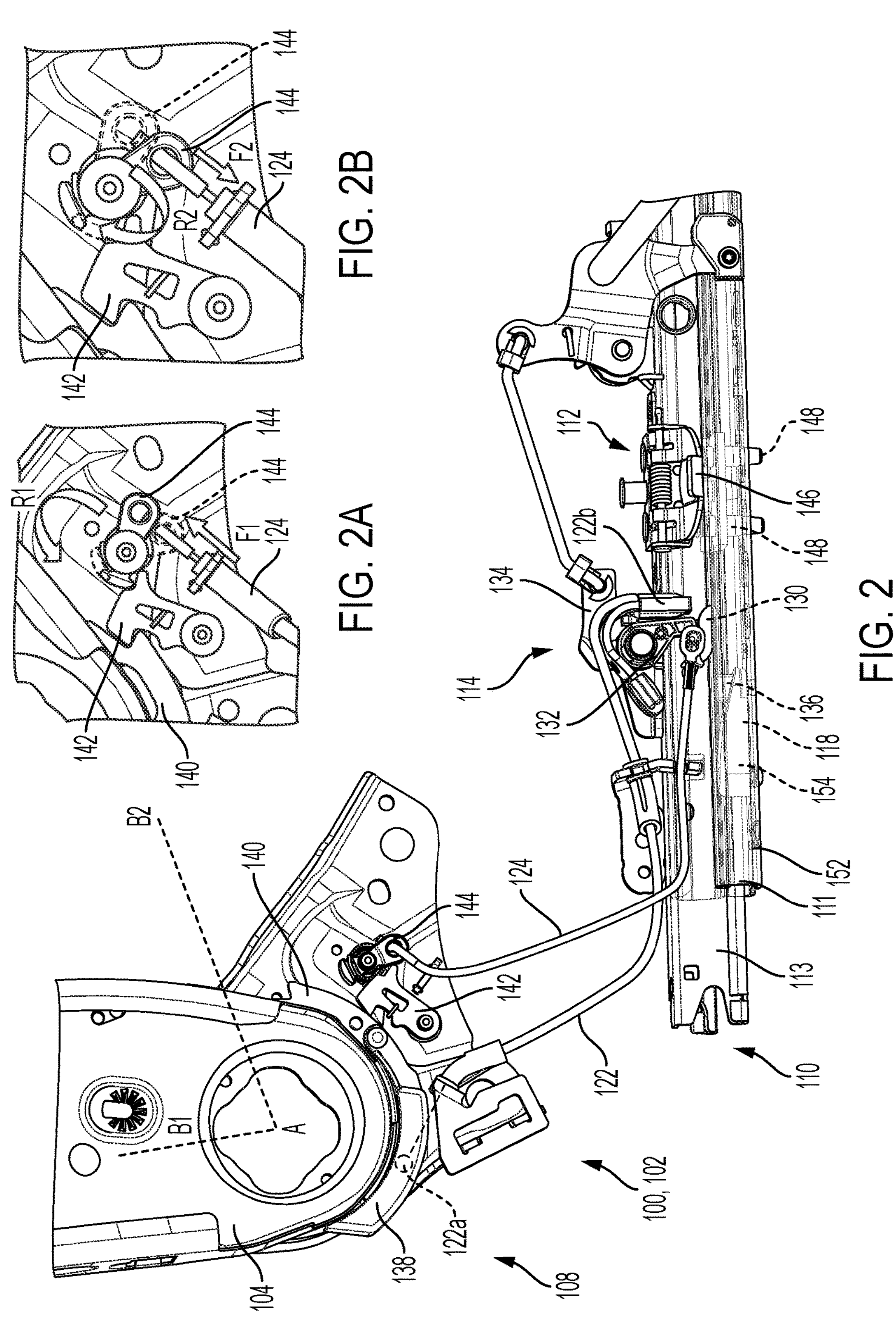

FIG. 2B

FIG. 2A

FIG. 2

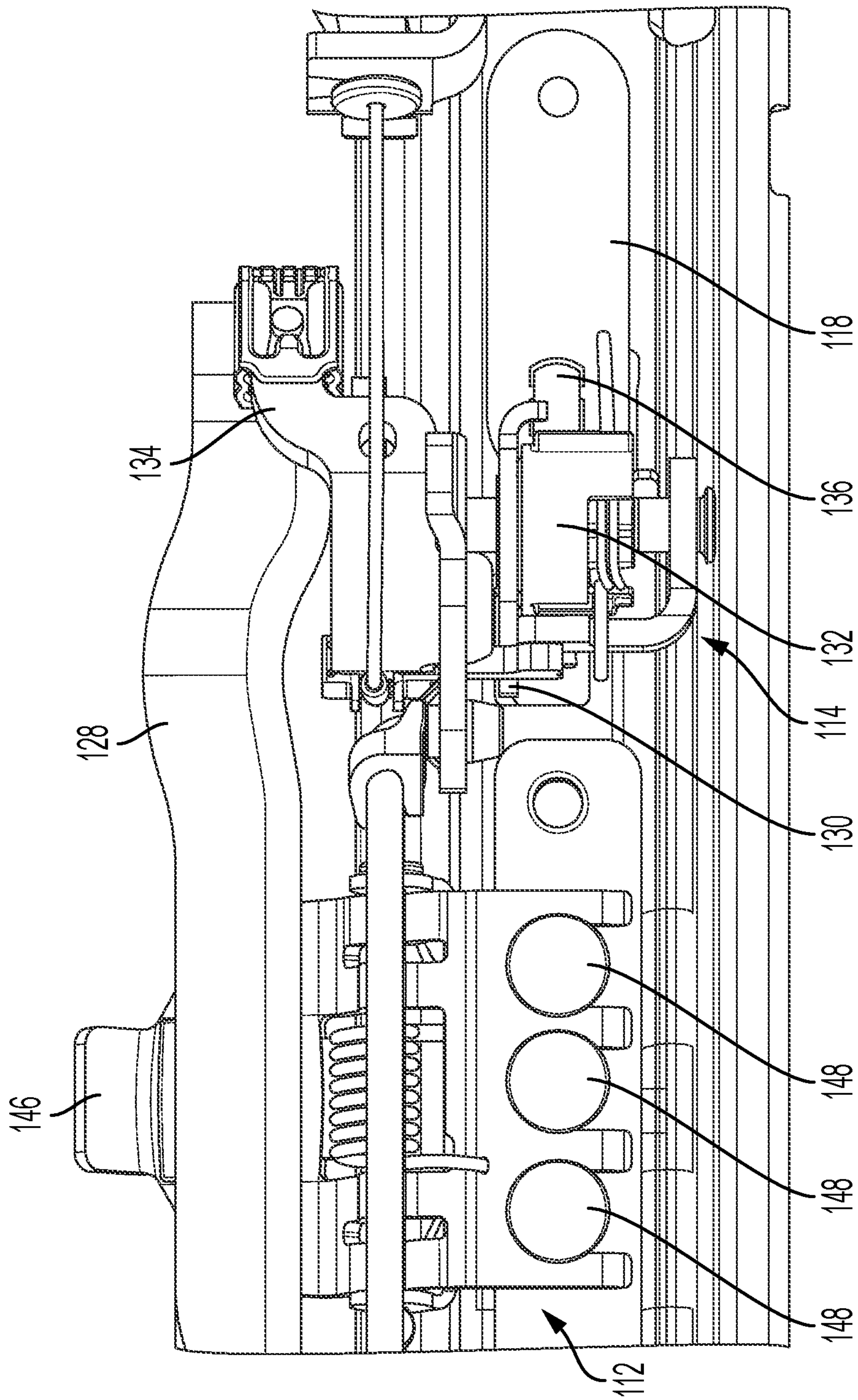

FIG. 3

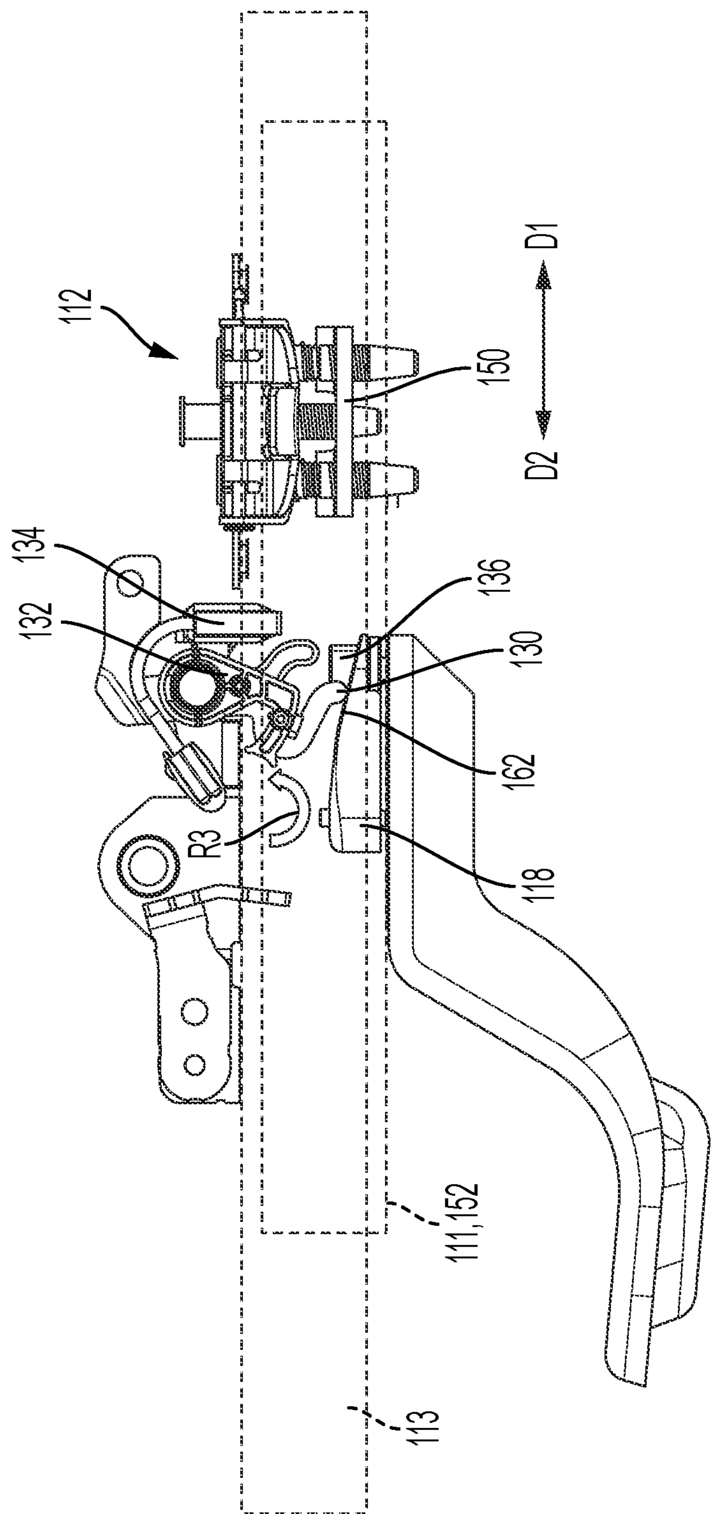

FIG. 4

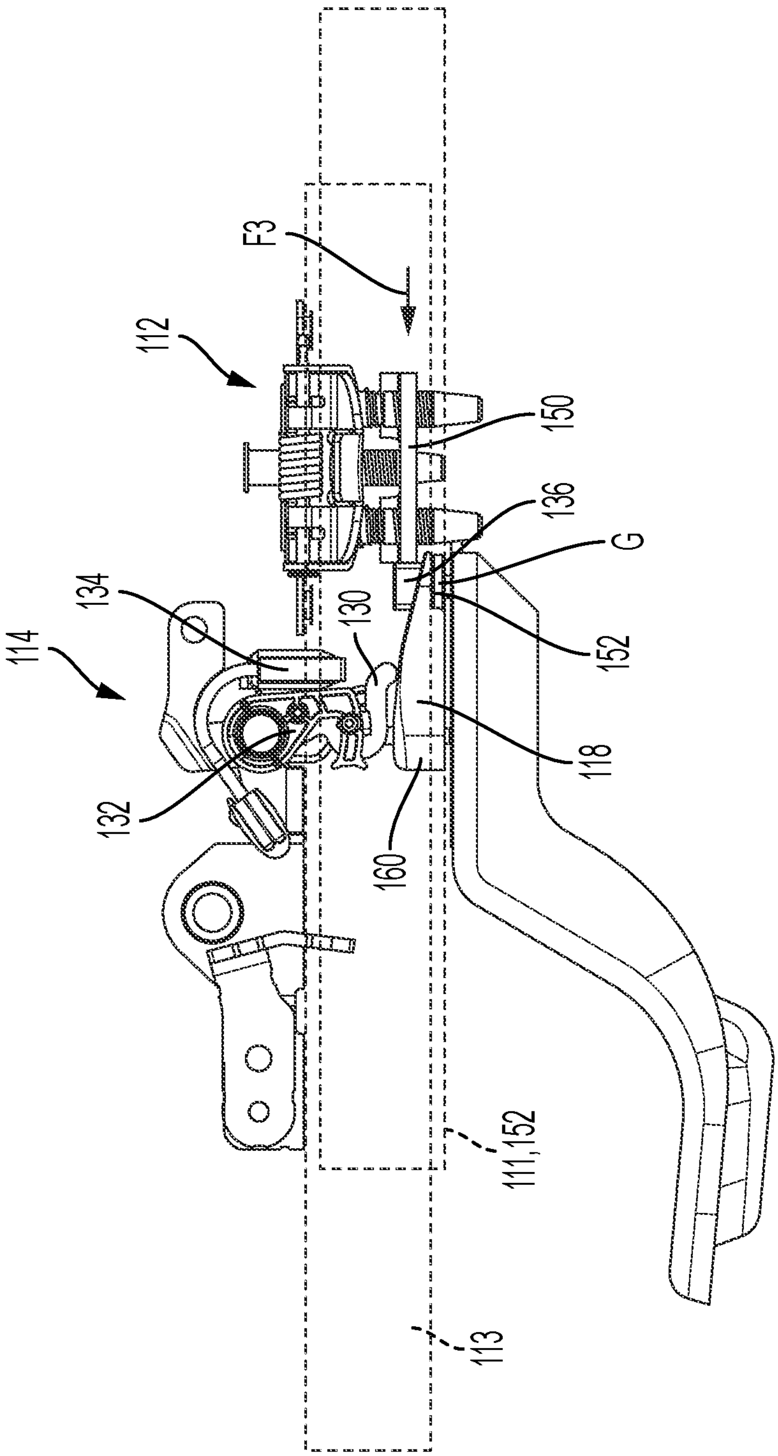

FIG. 5

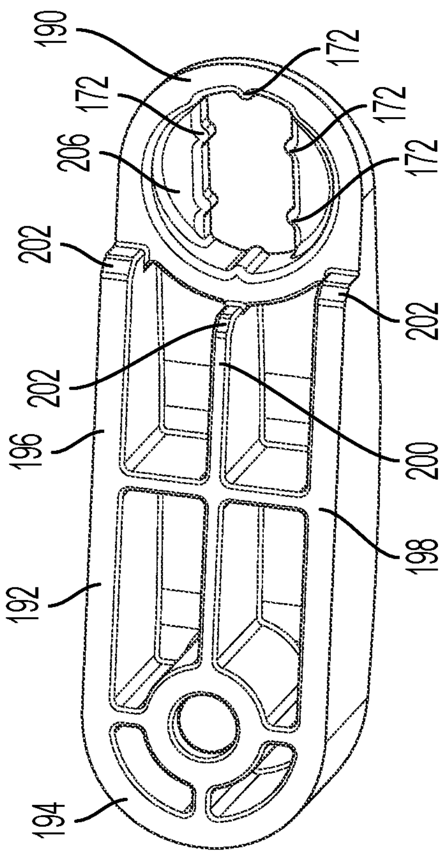
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FIG. 6B

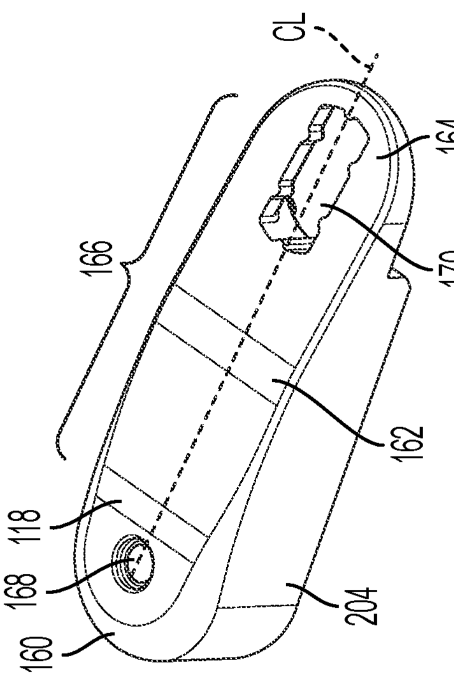
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FIG. 6A

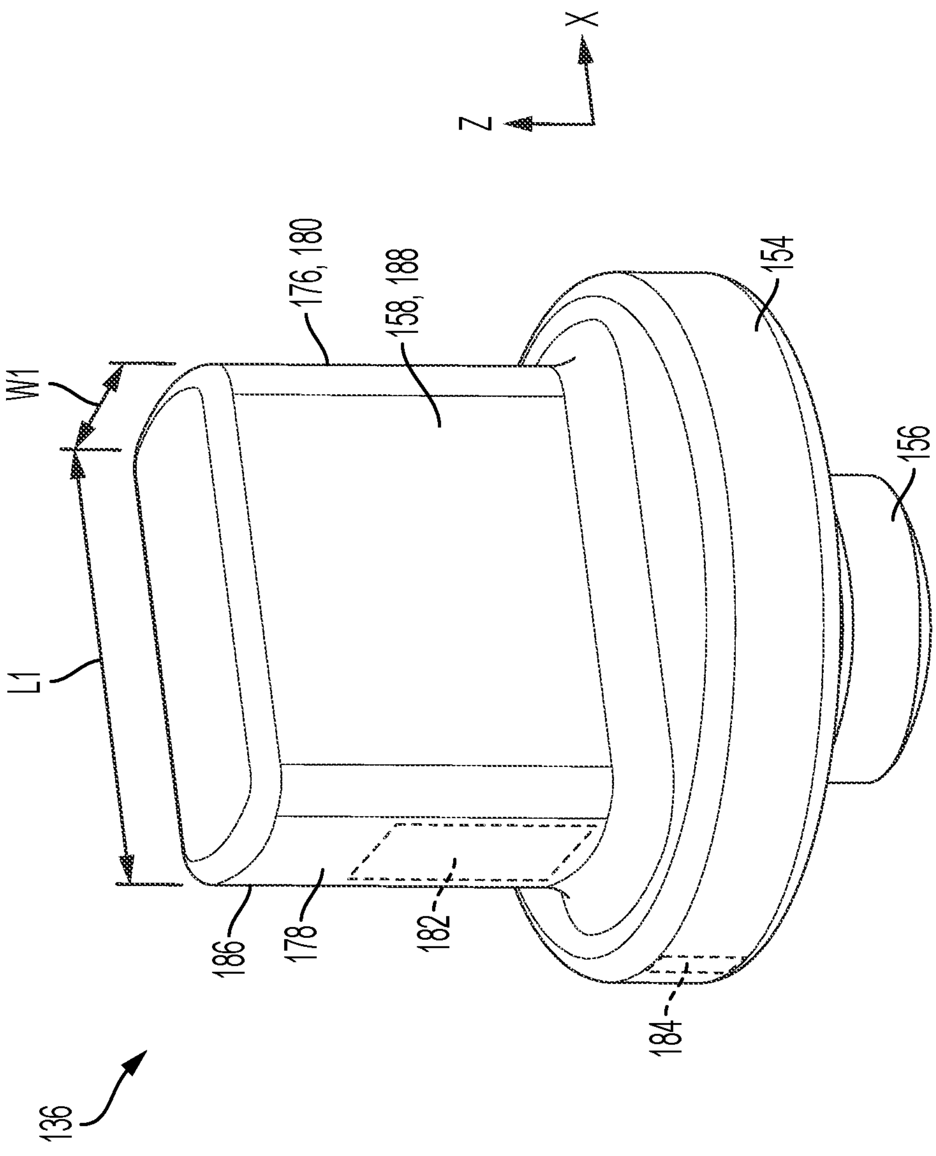

FIG. 7

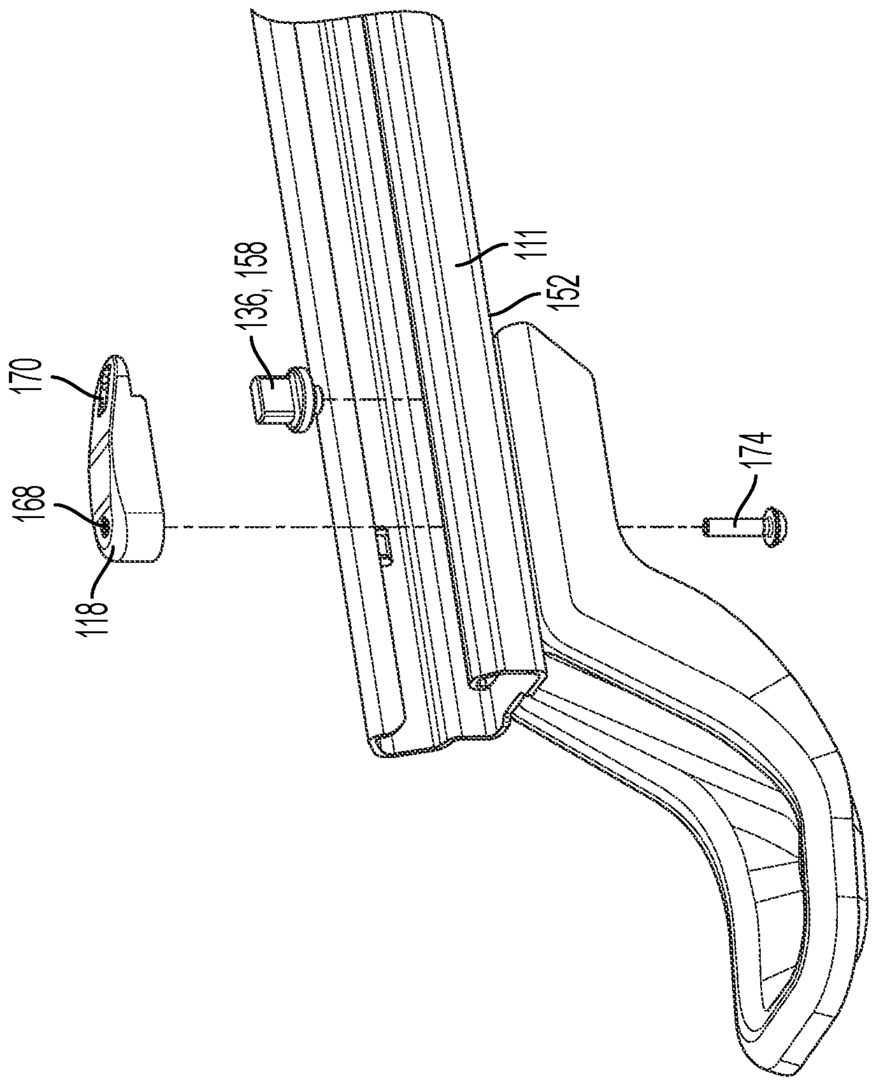

FIG. 8

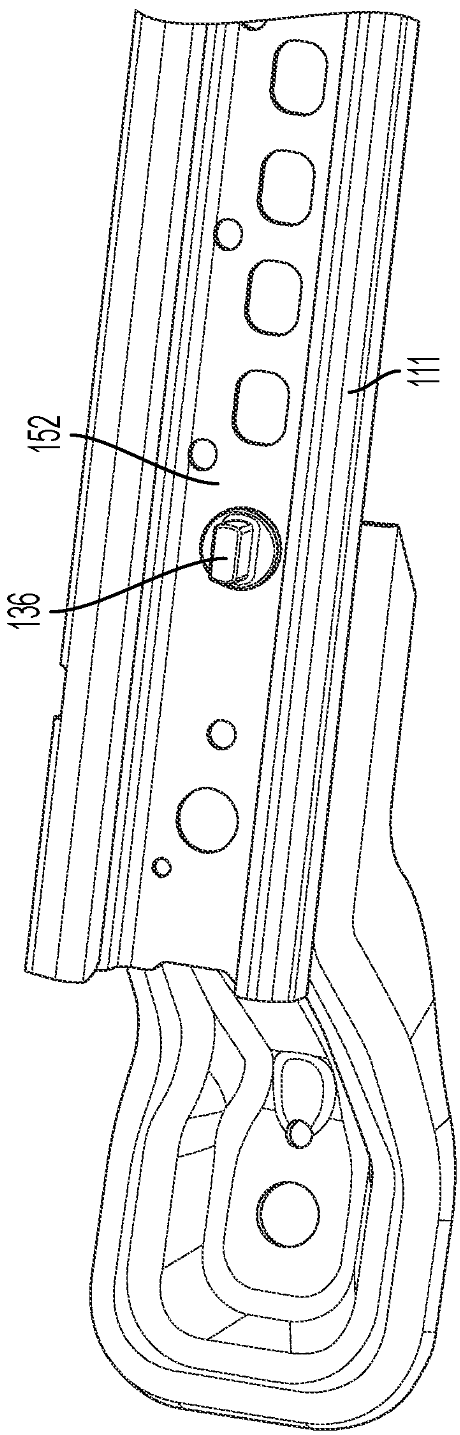

FIG. 9

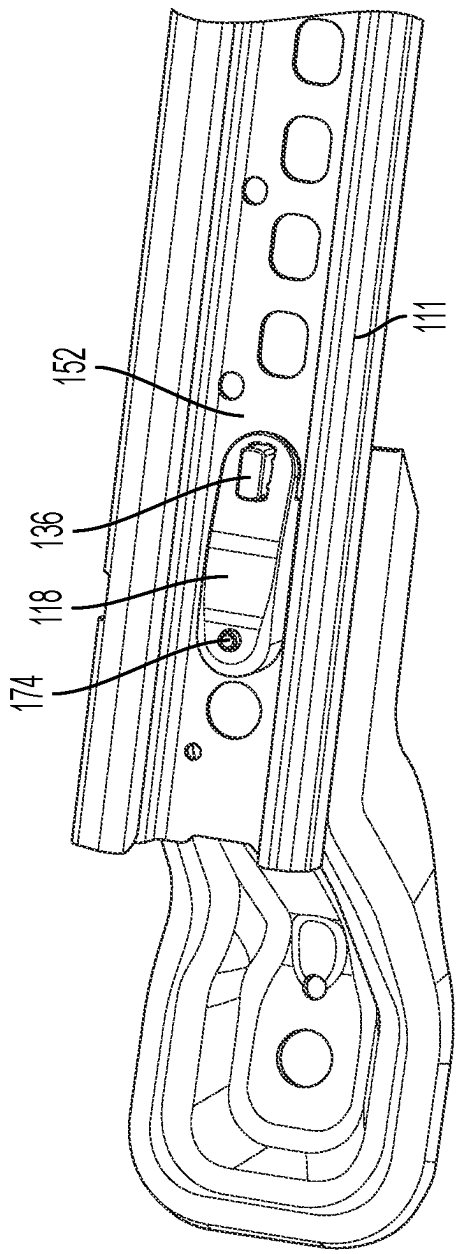

FIG. 10

VEHICLE SEAT ADJUSTMENT ASSEMBLY

TECHNICAL FIELD

[0001] The present disclosure relates to a vehicle seat, in particular a longitudinal adjustment assembly for use in vehicle seats.

BACKGROUND

[0002] A known adjusting system for a vehicle seat is described in U.S. Pat. No. 8,141,953.

SUMMARY

[0003] According to one embodiment, a vehicle seat is provided. The vehicle seat may include a rail assembly, a seat sub-frame, a backrest, an ingress-egress aid, and a memory ramp. The rail assembly may include a fixed rail, a translatable rail, and a locking device. The locking device may be configured to operate in a locked state, in which the locking device locks the translatable rail to the fixed rail, and an unlocked state in which the locking device unlocks the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail. The seat sub-frame may be fixed to the translatable rail and the backrest may be pivotally connected to the seat sub-frame. The backrest may be configured to pivot towards a seating surface of the seat sub-frame and the translatable rail may be configured to carry the seat sub-frame and the backrest. The ingress-egress aid may be coupled to the backrest and the locking device and in response to the backrest pivoting towards the seating surface, the ingress-egress aid may be configured to change the locking device from the locked state to the unlocked state to collectively permit the translatable rail to move along the fixed rail in a first direction to an ingress-egress position. The memory ramp may be fixed to the fixed rail, and the ingress-egress aid may include a mechanical effector. As the translatable rail moves in a second direction, opposite the first direction, from the ingress-egress position to one or more use positions, the mechanical effector may be configured to move along the memory ramp in the second direction and in a vertical direction to permit the backrest to pivot away from the seating surface.

[0004] According to another embodiment, a rail assembly for use in a vehicle seat is provided. The vehicle seat may include a seat sub-frame, a backrest, and an ingress-egress aid. The backrest may be pivotally connected to the seat sub-frame and configured to pivot towards a seating surface of the seat sub-frame and the ingress-egress mechanism may be coupled to the backrest. The rail assembly may include a fixed rail, a translatable rail, a locking device, and a memory bracket. The translatable rail may include a first stop member and may be configured to carry the seat sub-frame. The locking device may be configured to be coupled to the ingress-egress mechanism and the locking device may be further configured to operate in a locked state, in which the locking device locks the translatable rail to the fixed rail, and an unlocked state in which the locking device unlocks the translatable rail from the fixed rail to permit translation of the translatable rail in a longitudinal direction and along the fixed between one or more use positions to an ingress-egress position. The ingress-egress mechanism may be configured to, responsive to the backrest moving towards the seating surface, change the locking device from the locked state to the unlocked state. The memory bracket may be fixed to the fixed rail and may include a second stop member. When the translatable rail is in a use position of the one or more use positions, the first stop member may contact the second stop member. As the translatable rail moves in the longitudinal position from the ingress-egress position to the one or more use positions, a portion of the memory bracket may engage and displace a mechanical effector of the ingress-egress mechanism to permit the backrest to pivot away from the seating surface.

[0005] According to yet another embodiment, a vehicle seat is provided. The vehicle seat may include a rail assembly, a seat sub-frame, a backrest, an ingress-egress mechanism, a memory bracket, and a second stop member. The rail assembly may include a fixed rail, a translatable rail, and a locking device. The translatable rail may include a first stop member, and the locking device may be configured to operate in a locked state, in which the locking device locks the translatable rail to the fixed rail and an unlocked state in which the locking device unlocks to the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail. The seat sub-frame may be fixed to the translatable rail and the backrest may be pivotally connected to the seat sub-frame and configured to pivot towards a seating surface of the seat sub-frame. The translatable rail may be configured to carry the seat sub-frame and the backrest. The ingress-egress mechanism may be coupled to the backrest and the locking device, and in response to the backrest pivoting towards the seat surface, the ingress-egress mechanism may be configured to change the locking device from the locked state to the unlocked state to permit the translatable rail to move along the fixed rail in a first direction from one or more use positions to an ingress-egress position. The memory bracket may be fixed to the fixed rail and may form an inclined plane. As an example, the height of the inclined plane may increase with respect to a second direction, the second direction opposite the first direction. The second stop member may be coupled to the memory bracket and as the translatable rail is in a use position of the one or more use positions, the first stop member may contact the second stop member. The ingress-egress mechanism may include a mechanical effector. As the translatable rail moves in the second direction from the ingress-egress position to the one or more use positions, the inclined plane may be configured to engage the mechanical effector to permit the backrest to pivot away from the seating surface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates a schematic diagram of an exemplary vehicle seat according to one or more embodiments.

[0007] FIG. **2** illustrates a side view of a portion of a vehicle seat according to one or more embodiments.

[0008] FIGS. **2A-2B** illustrate side-detail views of a portion of a seat sub-frame of the vehicle seat shown in FIG. **2**.

[0009] FIG. **3** illustrates a top view of a rail assembly and a portion of an ingress-egress mechanism according to one or more embodiments.

[0010] FIG. **4** illustrates a side view of a portion of the rail assembly shown in FIG. **3** in which a translatable rail is moving to one or more use positions.

[0011] FIG. **5** illustrates a side view of the portion of the rail assembly shown in FIG. **4** in which the translatable rail is disposed in a use position, such as a full rear position.

[0012] FIG. **6**A illustrates a top-perspective view of a portion of a memory bracket according to one or more embodiments.

[0013] FIG. **6**B illustrates a bottom-perspective view of a portion of a memory bracket according to one or more embodiments.

[0014] FIG. **7** illustrates a perspective view of a stop member according to one or more embodiments.

[0015] FIG. **8** illustrates an exploded-perspective view of a fixed rail, the memory bracket and the stop member illustrated in FIG. **7** and FIG. **6**B, respectively.

[0016] FIG. **9** illustrates a perspective view of a fixed rail and the stop member illustrated in FIG. **7** and FIG. **6**B, respectively

[0017] FIG. **10** illustrates a perspective view of the fixed rail, the memory bracket, and the stop member illustrated in FIG. **8**.

DETAILED DESCRIPTION

[0018] Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative bases for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures can be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical application. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0019] "A", "an", and "the" as used herein refers to both singular and plural referents unless the context clearly dictates otherwise. By way of example, "a processor" programmed to perform various functions refers to one processor programmed to perform each and every function, or more than one processor collectively programmed to perform each of the various functions.

[0020] The term "substantially" or "about" may be used herein to describe disclosed or claimed embodiments. The term "substantially" or "about" may modify a value or relative characteristic disclosed or claimed in the present disclosure. In such instances, "substantially" or "about" may signify that the value or relative characteristic it modifies is within +0%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5% or 10% of the value or relative characteristic.

[0021] When an element or layer is referred to as being "on," "engaged to," "connected to," or "coupled to" another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). The term "and/or" includes any and all combinations of one or more of the associated listed items.

[0022] Although the terms first, second, third, etc. may be used to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0023] Spatially relative terms, such as "inner," "outer," "beneath," "below," "lower," "above," "upper," and the like, may be used for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" the other elements or features. Thus, the example term "below" can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0024] Certain vehicle seats, such as first row seats of a two door vehicle (e.g., coupe) and second or third row seats of a four door vehicle (e.g., cross-over, sport-utility vehicle) may include one or more easy-entry or ingress-egress mechanisms allowing the seat to be moved forward and downward (e.g., from a use position to a non-use position) and out of the way of a passenger entering (e.g., ingress) or exiting (e.g., egress) the vehicle. Once the passenger has entered or exited the vehicle, the passenger or another may move the seat back to the position (e.g., the use position), in which the seat was disposed prior to moving the seat. To enable relatively quick movement between the use and non-use positions, the vehicle seats are generally manually adjustable and require unlocking of a longitudinal adjustment assembly and the backrest prior to moving to the non-use position and the locking the same once in the use position.

[0025] To return from the non-use position to the use position, the user or passenger may push or pull on one or more portions of the seat so that the longitudinal adjustment assembly is actuated and returned to the use position. As the longitudinal adjustment systems is slid backwards, portions of the easy-entry or ingress-egress mechanisms may be triggered by a memory component (e.g., memory stone) disposed within the longitudinal adjustment assembly so that the backrest may return to the use position and portions of the longitudinal adjustment assembly may be stopped in the use position.

[0026] Under certain conditions, such as when a user or passenger slides the longitudinal adjustment assembly back to the use position with an excessive force, the memory component may break or become dislodged from its desired position. Breaking or dislodgement of the memory component may render the seat inoperable and cause customer dissatisfaction.

[0027] The present disclosure attempts to remedy the above-mentioned problems.

[0028] FIG. **1** illustrates a schematic view of an exemplary vehicle seat **100** according to one or more embodiments. The vehicle seat **100** may include a seat sub-frame **102** and a backrest **104** that may be pivotally connected to the seat sub-frame **102**, via a pivot mechanism **108**, so that the backrest **104** may pivot towards a seat surface **106** of the seat sub-frame **102**, about an axis A from a use position B1 to an ingress-egress position B2, as represented by hidden lines **104**. The vehicle seat **100** may include one or more rail assemblies **110** that may each include a fixed rail **111**, a translatable rail **113**, and a locking device **112** that may be configured to operate in a locked state, in which the translatable rail **113** is locked or fixed to the fixed rail **111**, and an unlocked state in which the translatable rail **113** may translate in a longitudinal direction (e.g., as represented by the x-axis).

[0029] The vehicle seat **100** may further include an ingress-egress mechanism **114** that may be operatively coupled to the locking device **112** and the pivot mechanism **108** by one or more cables, such as an activation cable **122** and a communication cable **124**. A known ingress-egress mechanism is described in U.S. Pat. No. 8,141,953 which is hereby incorporated by reference.

[0030] The pivot mechanism **108** may be operatively connected to a strap **120** by a backrest cable **126**, so that as a user pulls the strap **120**, the pivot mechanism **108** actuates to pivot the backrest **104** (e.g., as represented by the bi-directional arrows) towards the seating surface **106** and to unlock the locking device **112** so that the backrest **104** and the seat sub-frame **102** is movable (e.g., in a first or forward direction that is substantially parallel to the x-axis) to an ingress-egress position. When the seat **100** is in the ingress-egress position, the seat **100** may be positioned to provide additional space between the vehicle seat **100** and portions of a vehicle opening (e.g, B-pillar, C-pillar) or another vehicle seat (e.g., second row or third row).

[0031] The seat **100** may also include a release handle or lever **128** that may be operatively connected to the locking device **112**, so that as a user applies a force to the handle **128** (e.g., a vertical force as represented by the z-axis) the locking device may change to the unlocked state. In one or more embodiments, the rail assembly **110** may include a memory bracket **118** that may be fixed to the fixed rail **111**. When the translatable rail **113**, which carries the seat sub-frame **102** and the backrest **104**, is moved from the non-use position (e.g., in a second direction, opposite the first direction) one or more components of the ingress-egress mechanism **114** may engage and move along portions of the memory bracket **118** to permit the backrest **104** to move from the ingress-egress position B2 to the use position B1.

[0032] FIG. **2** illustrates a side-perspective view of portions of the vehicle seat **100**. For purposes of clarity, portions of the vehicle seat **100** are not shown, those portions include the seating surface **106**, release handle or lever **128**, and pivot mechanism **108**, among other components. FIG. **3** illustrates a top view of a portion of a rail assembly **110**, a portion of the release lever **128**, and a portion of the ingress-egress mechanism **114**. FIG. **4** illustrates a side view of the rail assembly **110** in which the translatable rail **113** is moving in the second direction towards one or more use positions. FIG. **5** illustrates a side view of the rail assembly **110** in which the translatable rail **113** is disposed in one or more of the use positions (e.g., a full rear position). For purposes of clarity, the fixed rail **111** and the translatable rail **113** are represented by hidden lines in FIG. **4** and FIG. **5**, and a number of components shown and described in FIG. **2** are not shown.

[0033] As the backrest **104** pivots about the axis A from the use position B1 to the non-use position B2, a backrest actuation bracket **138** fixed to the backrest **104** may pivot with the backrest **104** to pull or actuate the activation cable **122**. The activation cable **122** may include a first end **122*a*** that may be fixed to the backrest actuation bracket **138** and a second end **122*b*** that may be fixed or coupled to an activation lever **134** of the ingress-egress mechanism **114**. An end of the activation lever **134** may engage an end of the release handle or lever **128** so that as the activation cable **122** is pulled by movement of the backrest actuation bracket **138**, the activation lever **134** may pivot and displace the release handle **128**. Pivoting of the release handle **128** may engage and rotate the locking lever **146** of the locking device **112**, to lift the pins **148** and change the state of the locking device from the locked state to the unlocked state.

[0034] The seat sub-frame **102** may include one or more components configured to lock or latch the backrest **104** in the non-use position B2. As an example, a catch **142** may be configured to engage a backrest locking bracket **140** that may be fixed to the backrest **104**, the backrest actuation bracket **138**, or both. In one or more embodiments, a pawl **144** may be provided to prevent the catch **142** from disengaging the backrest locking bracket **140**. The pawl **144** may be pivotally connected to one or more portions of the seat sub-frame **102** and operatively connected to the communication cable **124**. As will be described in greater detail below, the catch **142**, the pawl **144**, and the locking bracket **140** may be collectively configured to prevent the backrest **104** from moving to the use position from the non-use position or ingress-egress position B2 until the translatable rail **113** has returned to one or more of the use positions (e.g., a full rear position).

[0035] A first end of the communication cable **124** may be fixed to the pawl **144** and a second end of the communication cable **124** may be operatively connected to one or more components of the ingress-egress mechanism **114** such as a mechanical effector or use position lever **130**. As an example, the ingress-egress mechanism **114** may include a coupling lever **132** that is coupled to the use position lever **130** and the second end of the communication cable **124** may be fixed to the coupling lever **132**. Additionally or alternatively, the coupling lever **132** may be biased by one or more springs so as the translatable rail **113** and backrest **104** move or are moved to the ingress-egress positions, a force F1 is applied by the communication cable **124** to rotate the pawl **144** to the engage the catch **142**.

[0036] As the translatable rail **113** is moved in the second direction D2 towards the use position, the use position lever **130** may engage and directly contact one or more portions of the memory bracket **118** to move (e.g., rotate about rotational direction R3) the use position lever **130** in the second direction D2 (e.g., by way movement of the translatable rail **113**) and the vertical direction (e.g., away from a bottom wall **152** of the fixed rail **111**). Movement of the use position lever **130** may be transmitted through the coupling lever **132** and the communication cable **124** so that a second force F2 is applied to the pawl **144** thereby causing the pawl **144** to rotate (e.g., in the second rotational direction R2) away from the catch **142** and allowing the backrest **104** to return of the use position B1.

[0037] FIG. **6**A and FIG. **6**B show a top-perspective view and a bottom-perspective view, respectively of the memory bracket **118** according to one or more embodiments. The memory bracket **118** may be positioned to engage the use position lever **130** as the translatable rail **111** moves from the ingress-egress position to the use position. In one or more embodiments, a first stop member **150** such as a locking device bracket may be fixed to the translatable rail **113** so that when the translatable rail **113** is disposed in one or more of the use positions (e.g., full rear), the first stop member **150** directly contacts a second stop member **136** (FIG. **5**) that may be fixed to one or more portions of the fixed rail **111**. As an example, the second stop member **136** may be coupled to the memory bracket **118** or be integrally formed with the memory bracket **118**.

[0038] In one or more embodiments, the second stop member **136** may be formed by a fastener (e.g., a rivet) provided with a shaft or shank **156** that may extend through the bottom wall **152** of the fixed rail **111**. The second stop member **136** may also include a collar **154** and a stop post **158**, the collar **154** may be disposed between the stop post **158** and the shank **156**. The stop post **158** may shaped, positioned, and/or sized so that as the mechanical effector or the use position lever **130** moves along portions of the memory bracket **118**, neither the use position lever **130** nor any other portion of the ingress-egress mechanism contacts the stop post **158** or any other portion of the second stop member **136**.

[0039] As illustrated in FIG. **7**, the stop post **158** may have a substantially cylindrical body (e.g., rectangular) shape and the second stop member **136** may include a front side or first side **176**, a rear side or second side **178**, and third and fourth sides **186**, **188** or transverse sides extending between the first side **186** and the second side **188**. As an example, the first side or front side **176** may face in the first direction D1 and the second side **178** or rear side may face in the second direction D2, and the first and second sides **176**, **178** may have a first width W1. The lateral sides or third and fourth sides **186**, **188** may have a first length L**2** that may be greater than the first width W1.

[0040] The first side **176** may include a first surface **180** that is positioned to engage the first stop member or stop bracket **150** when the translatable rail **113** is in one or more of the use positions (e.g., full rear, as illustrated in FIG. **5**) and the second side **178** may include a second surface **182** and a third surface **184** that may be spaced apart from one another with respect to the vertical direction (e.g., as represented by the z-axis). The second surface **182** and the third surface **184** may be configured to engage portions of the memory bracket **118** so that load applied from the first stop member or stop bracket **150** sliding into the second stop member or end stop **136** is transferred and at least partially dissipated by and through the end stop **136**, through the memory bracket **118**, and into the fastener **174**.

[0041] The memory bracket **118** may include a rear portion **160**, a front portion **164**, and a medial or effective portion **162** that is disposed between the front and rear portions **160**, **164**. The effective portion **162** may be the portion that engages the mechanical effector or use position lever **130** so that the use position lever **130** is actuated (FIG. **4**) to permit the backrest **104** to move from the non-use position to the use position. The memory bracket **118** may taper in the vertical direction from the rear portion **160** to the front portion **164** to form an inclined plane **166** extending from the rear portion **160** to the front portion **164**. The rear portion **160** of the memory bracket **118** may define one or more fastener apertures **168** that may be configured to receive a fastener **174** that may extend through the bottom wall **152** of the fixed rail **111** to fix the rear portion **160** of the memory bracket **118** to the fixed rail **111**.

[0042] The front portion **164** may define a stop member aperture **170** configured to receive portions (e.g., the stop post **158** of the second stop member **136**) so that the stop post **158** extends through the stop member aperture **170**. The stop member aperture **170** may have an elongated shape and be disposed along a center CL of the memory bracket **118**. The fastener aperture **168** may be disposed along the center CL of the memory bracket **118** so that forces applied to the second end stop member **136** may be transmitted in a relatively straight direction towards the fastener **174**. An inner periphery of the stop member aperture **170** may include one or more locking protrusions **172** that extend into the aperture and engage the stop post **158** to form a force-fit or press-fit condition between the memory bracket **118** and the stop post **158**. In one or more embodiments, lateral sections **206** of the front portion **164** that define the stop member aperture **170** may be recessed with respect to the bottom portion **190** to form a pocket, so that portions of the collar **154** nest within the pocket and abut against the lateral sections **206**. The pocket and lateral sections **206** may facilitate positioning the second stop member **136** with respect to the memory bracket **118** during assembly and may also facilitate load distribution when the translatable rail **113** and first stop member **150** slide into the second stop member **136**.

[0043] A bottom surface **190** of the front portion **164** may be offset with respect to a bottom surface **192** of the effective or medial portion **162** and/or a bottom surface **194** of the rear portion **160** so that the bottom surface **190** of the front portion **164** is spaced apart from the bottom wall **152** to form a gap G and the collar **154** may be disposed within the gap G. Because the collar **154** is disposed within the gap G, the third surface **184** may transfer the force F3 to a bottom portion of the mounting bracket **118** in addition to the force transferred by the second surface **182**.

[0044] The bottom surfaces **190**, **192**, **194** may collectively form a portion of a base **204** of the memory bracket **118**, and the base **204** may include one or more arms including a first arm **196**, a second arm **198**, and a medial arm **200**. The first and second arms **196**, **198** may each extend along first and second edge regions and each of the arms **196**, **198**, **200** may include distal end portions **202**. The distal end portion **202** of the medial arm **200** may be configured to engage the third surface **184** of the collar **154** and the distal end portions **202** of the first and second arms **196**, **198** may engage other surfaces of the collar **154** to rotationally fix the memory bracket **118** to the bottom wall **152** of the fixed rail **111**.

[0045] FIG. **8** illustrates an exploded view of the memory bracket **118**, second stop member **136**, and the fastener **174** being assembled to the fixed rail **111**. FIG. **10** illustrates the memory bracket **118**, second stop member **136**, and the fastener **174** assembled to the fixed rail **111**. During assembly, the second stop member **136** (e.g., the shank **156** shown in FIG. **7**) may be inserted into an aperture defined by the bottom wall **152** of the fixed rail. The memory bracket **118** may be positioned and placed upon the bottom wall **152** so that the stop post **158** extends through the stop member aperture **170** and the fastener **174** may be inserted from and through the bottom wall **152** of the fixed rail **111** and into the fastener aperture **168**.

[0046] FIG. **9** illustrates a top-perspective view of the second stop member **136** fixed or installed to the fixed rail **111**. Under certain circumstances, such as when vehicle seat does not include an ingress-egress mechanism **114**, the second stop member **136** may be used without the memory bracket. In this configuration, the collar **154** may lie along the bottom wall of the fixed rail.

[0047] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, to the extent any embodiments are described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics, these embodiments are not outside the scope of the disclosure and can be desirable for particular applications.

[0048] The following is a list of reference numbers shown in the Figures. However, it should be understood that the use of these terms is for illustrative purposes only with respect to one embodiment. And, use of reference numbers correlating a certain term that is both illustrated in the Figures and present in the claims is not intended to limit the claims to only cover the illustrated embodiment.

PARTS LIST

[0049] **100** vehicle seat
[0050] **102** seat sub-frame
[0051] **104** backrest
[0052] **106** seating surface
[0053] **108** pivot mechanism
[0054] **110** rail assembly
[0055] **111** fixed rail
[0056] **112** locking device
[0057] **113** translatable rail
[0058] **114** ingress-egress mechanism
[0059] **116** vehicle floor
[0060] **118** memory bracket
[0061] **120** strap
[0062] **122** activation cable
[0063] **122***a* first end of activation cable
[0064] **122***b* second end of activation cable
[0065] **124** communication cable
[0066] **126** backrest cable
[0067] **128** release handle and lever
[0068] **130** effector, use position lever
[0069] **132** effector, coupling lever
[0070] **134** activation lever
[0071] **136** second stop member, end stop
[0072] **138** backrest actuation bracket
[0073] **140** backrest locking bracket
[0074] **142** catch
[0075] **144** pawl
[0076] **146** locking lever
[0077] **148** pins
[0078] **150** stop bracket
[0079] **152** bottom wall of fixed rail
[0080] **154** collar of second stop member
[0081] **156** shank of second stop member
[0082] **158** stop post
[0083] **160** rear portion
[0084] **162** effective portion
[0085] **164** front portion
[0086] **166** inclined plane
[0087] **168** fastener aperture
[0088] **170** stop member aperture
[0089] **172** locking protrusion
[0090] **174** fastener
[0091] **176** first side, front side
[0092] **178** second side, rear side
[0093] **180** first surface
[0094] **182** second surface
[0095] **184** third surface
[0096] **186** third side, lateral side
[0097] **188** fourth side, lateral side
[0098] **190** bottom surface front portion
[0099] **192** bottom portion effective portion
[0100] **194** bottom portion rear portion
[0101] **196** first arm, edge region
[0102] **198** second arm, edge region
[0103] **200** third arm, medial region
[0104] **202** distal end portions
[0105] **204** base memory bracket
[0106] **206** recessed surface

What is claimed is:

1. A vehicle seat comprising:

a rail assembly including a fixed rail, a translatable rail, and a locking device, the locking device configured operate in a locked state in which the locking device locks the translatable rail to the fixed rail, and an unlocked state, in which the locking device unlocks the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail;

a seat sub-frame fixed to the translatable rail;

a backrest pivotally connected to the seat sub-frame and configured to pivot towards a seating surface of the seat sub-frame, wherein the translatable rail is configured to carry the seat sub-frame;

an ingress-egress aid coupled to the backrest and the locking device, wherein in response to the backrest pivoting towards the seat surface, the ingress-egress aid is configured to change the locking device from the locked state to the unlocked state to collectively permit the translatable rail to move along the fixed rail in a first direction to an ingress-egress position; and a memory ramp fixed to the fixed rail, wherein the ingress-egress aid includes a mechanical effector, wherein as the translatable rail moves in a second direction from the ingress-egress position to one or more use positions, the mechanical effector moves along the memory ramp in the second direction and a vertical direction to permit the backrest to pivot away from the seating surface, and wherein the second direction is substantially opposite to the first direction.

2. The vehicle seat of claim **1**, further comprising:

a stop bracket coupled to and configured to translate with the translatable rail; and

an end stop extending through a portion of the memory ramp and fixed to the fixed rail, wherein the one or more use positions includes a full rear position, in which the stop bracket directly contacts the end stop.

3. The vehicle seat of claim **2**, wherein the stop bracket does not engage the memory ramp.

4. The vehicle seat of claim **1**,

a stop bracket coupled to and configured to translate with the translatable rail; and

an end stop, wherein the one or more use positions includes a full rear position, in which the stop bracket directly contacts the end stop,

wherein the memory ramp includes a rear portion, a front portion, and an effective portion extending between the rear portion and the front portion, wherein the mechanical effector is configured to engage the effective portion to permit the backrest to pivot away from the seating surface, and

wherein the end stop extends through the front portion and is fixed to the fixed rail.

5. The vehicle seat of claim **4**, further comprising:

a fastener, wherein the rear portion defines an aperture and the fastener extends through the fixed rail and into the aperture to fix the memory ramp to the fixed rail.

6. The vehicle seat of claim **4**, wherein the a bottom surface of the front portion is recessed with respect to a bottom surface of at least one of the rear portion and the effective portion, wherein the bottom surface of the front portion and a portion of the fixed rail form a gap,

wherein the end stop includes a collar and a stop post extending from the collar, wherein the stop bracket is configured to directly contact the stop post and the collar is disposed within the gap.

7. The vehicle seat of claim **1**,

a stop bracket coupled to and configured to translate with the translatable rail; and

an end stop including a stop post, wherein when the translatable rail is disposed in a use position of the one or more use positions the stop bracket directly contacts the stop post, and

wherein the stop post and the mechanical effector are collectively so that as the mechanical effector moves along the memory ramp in the second direction, the mechanical effector does not contact the stop post of the end stop.

8. The vehicle seat of claim **7**, wherein the stop post has a substantially cylindrical body, the cylindrical body including a first transverse wall, a second transverse wall, a front wall, and a rear wall, wherein the first transverse wall and the second transverse wall oppose one another and the front wall and the rear wall each extend between the first transverse wall and the second transverse wall, wherein the first transverse wall has a first length and the front wall has a first width, and wherein the first length is greater than the first width.

9. The vehicle seat of claim **8**, wherein the stop post and the mechanical effector are collectively so that as the mechanical effector moves along the memory ramp in the second direction, a distal end of the mechanical effector is spaced apart from and disposed closer to the first transverse wall than the second transverse wall.

10. The vehicle seat of claim **8**, wherein the one or more use positions includes a full rear position, in which the stop bracket directly contacts the front wall of the stop post.

11. A rail assembly for use in a vehicle seat, the vehicle seat including a seat sub-frame, a backrest pivotally connected to the seat sub-frame and configured to pivot towards a seating surface of the seat sub-frame, and an ingress-egress aid coupled to the backrest, the rail assembly comprising:

a fixed rail;

a translatable rail configured to carry the seat sub-frame, wherein the translatable rail includes a first stop member;

locking device, the locking device configured to be coupled to the ingress-egress aid, wherein the locking device is further configured to operate in a locked state, in which the locking device locks the translatable rail to the fixed rail, and an unlocked state in which the locking device unlocks the translatable rail from the fixed rail to permit translation of the translatable rail in a longitudinal direction along the fixed rail between the one or more use positions to an ingress-egress position, wherein the ingress-egress aid is configured to, responsive to the backrest moving towards the seating surface, change the locking device from the locked state to the unlocked state; and

a memory bracket fixed to the fixed rail and including a second stop member, wherein when the translatable rail is in a use position of the one or more use positions, the first stop member contacts the second stop member, and

wherein as the translatable rail moves in the longitudinal direction from the ingress-egress position to the one or more use positions, a portion of the memory bracket engages and displaces a mechanical effector of the ingress-egress aid to permit the backrest to pivot away from the seating surface.

12. The rail assembly of claim **11**, wherein the memory bracket includes a rear portion, a front portion, and an effective portion extending between the front and rear portions, wherein the effective portion is the portion that engages and displaces the mechanical effector of the ingress-egress aid, and wherein a height of the memory bracket tapers from the rear portion to the front portion.

13. The rail assembly of claim **12**, wherein the front portion defines an aperture and the second stop member includes a stop post and a shank, wherein the stop post extends through the aperture defined by the front portion of the memory bracket and the shank extends through an aperture defined by a base of the fixed rail.

14. The rail assembly of claim **13**, wherein an inner periphery of the aperture defines one or more protrusions configured to engage one or more portions of the stop post

15. The rail assembly of claim **11**, wherein the second stop member includes a stop post, and wherein the stop post is integral with the memory bracket.

16. The rail assembly of claim **11**, further comprising:

a fastener, wherein the memory bracket includes a rear portion, a front portion, and an effective portion extending between the front and rear portions, wherein the effective portion is the portion that engages and displaces the mechanical effector of the ingress-egress aid, wherein the rear portion defines a fastener aperture and the fastener extends through the fixed rail and into the fastener aperture, and

wherein the fastener and the second stop member are substantially aligned with one another with respect to the longitudinal direction.

17. A vehicle seat comprising:

a rail assembly including a fixed rail, a translatable rail, and a locking device, wherein the translatable rail includes a first stop member, wherein the locking device is configured operate in a locked state, in which the locking device locks the translatable rail to the fixed rail, and an unlocked state in which the locking device unlocks the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail;

a seat sub-frame fixed to the translatable rail;

a backrest pivotally connected to the seat sub-frame and configured to pivot towards a seating surface of the seat sub-frame, wherein the translatable rail is configured to carry the seat sub-frame and the backrest;

an ingress-egress aid coupled to the backrest and the locking device, wherein in response to the backrest pivoting towards the seat surface, the ingress-egress aid is configured to change the locking device from the locked state to the unlocked state to permit the translatable rail to move along the fixed rail in a first direction from one or more use positions to an ingress-egress position;

a memory bracket fixed to the fixed rail, wherein the memory bracket forms an inclined plane, wherein a height of the inclined plane increases with respect to a second direction, wherein the second direction is opposite the first direction; and

a second stop member coupled to the memory bracket, wherein when the translatable rail is in a use position of the one or more use positions, the first stop member contacts the second stop member,

wherein the ingress-egress aid includes a mechanical effector, wherein as the translatable rail moves in the second direction from the ingress-egress position to the one or more use positions, the inclined plane engages and displaces the mechanical effector to permit the backrest to pivot away from the seating surface.

18. The vehicle seat of claim **17**, wherein the second stop member is formed by a fastener including a first side, facing the first direction, and a second side facing the second direction, wherein a first surface disposed on the first side is configured to directly contact the first stop member to stop the translatable rail in the use position, wherein the second side includes a second surface and a third surface, wherein the second and third surfaces are spaced apart from one another with respect to a vertical direction and wherein the first, second and third surfaces are collectively configured to transfer a force associated with stopping the translatable rail in the use position to the memory bracket.

19. The vehicle seat of claim **18**, wherein the third surface is spaced apart from the second surface with respect to the second direction.

20. The vehicle seat of claim **17**, wherein the second stop member includes a collar and a stop post extending from the collar, wherein the memory bracket includes a base configured to lie along a bottom wall of the fixed rail, wherein the base includes a first edge region and a second edge region, and wherein distal end portions of the first and second edge regions are each configured to engage the collar to rotationally fix the memory bracket to the bottom wall of the fixed rail.

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